# BENCHCOMPASS: From Scores to Signals for Training and Harness Decisions in Payment-Domain LLMs

**Sijie Dong**[*], **Wei Ren**[*], **Xuanwei Hu**, **Jiawei Luo**, **Zifan Wang**, **Xiaoyun Feng**, **Hui Cai**, **Lyuxin Xue**, **Peng Lu**, **Wei Wu**[†], **Xin Zhang**[†]
Ant International
{dongsijie.dsj,modi.rw,congyue.ww,evan.zx}@ant-intl.com

## Abstract

Payment operations are a critical financial infrastructure, but the value of large language models in this domain remains unclear because payment rules change quickly, evidence is fragmented, and decisions depend on transaction state, participant role, region, and payment rail. Existing benchmarks do not isolate whether failures come from missing payment-rule knowledge, poor use of supplied evidence, or brittleness under imperfect harness inputs. We introduce BENCHCOMPASS, a payment-domain benchmark whose construction pipeline builds scenario-grounded tasks from typed evidence packs, applies LLM-based quality checks, creates task-input attack variants, and reserves final item admission for domain experts. The release contains an expert-reviewed Pro benchmark covering payment knowledge, context-grounded scenario reasoning, and Attacked Open robustness, plus a lower-assurance Normal pool for inspection and future curation. Across 16 model variants, BENCHCOMPASS shows qualitatively different failure modes: missing parametric payment knowledge, incomplete reasoning over supplied rules, and failure to reject plausible but invalid workflows. The benchmark remains unsaturated: the best frontier model reaches 89.6% on Open Context-Grounded Reasoning and 81.7% under attacked inputs, while a representative 32B open-weight model reaches 69.8% and 42.6%. The benchmark, data, and code are publicly available at https://github.com/ant-intl/BenchCompass.

## 1 Introduction

Large language models have made rapid progress on broad and specialized evaluations, from multi-task knowledge in MMLU (Hendrycks et al., 2021) and diverse task coverage in BIG-bench (Srivastava et al., 2023) to expert scientific reasoning in GPQA (Rein et al., 2023). They also show strong capabilities in code generation benchmarks such as HumanEval (Chen et al., 2021a), dynamic benchmarks such as LiveBench (White et al., 2025), and long-context evaluations such as LongBench v2 (Bai et al., 2025). Payment is central to financial activity, and it is a complex domain rather than a static knowledge base or a single product. Payment systems are financial market infrastructures whose rules coordinate participants, procedures, and settlement obligations; these rules change across products, jurisdictions, rails, and compliance regimes, and payment decisions often depend on a transaction's state, timing, participant role, and rail-specific constraints. Existing financial benchmarks cover broad finance tasks (Xie et al., 2024), filing-grounded QA (Islam et al., 2023), and numerical or table-based financial reasoning (Chen et al., 2021b; Zhu et al., 2021; Chen et al., 2022), but not this fragmented and stateful payment-infrastructure setting.

Together, these properties make it difficult to assess what current LLMs can and cannot do in payment operations. This uncertainty matters for both communities: AI researchers lack evidence about which payment capabilities current models still miss, while payment practitioners cannot determine whether existing LLMs can materially support real workflows.

In this work, we ask whether LLMs can solve payment-domain tasks that require domain knowledge, evidence-grounded reasoning, and reliability under imperfect payment-knowledge harness engineering. Existing benchmarks usually report aggregate performance rather than separating miss-

[*] Equal contribution.
[†] Corresponding authors: congyue.ww@ant-intl.com and evan.zx@ant-intl.com.



*Findings of the Association for Computational Linguistics: EMNLP 2026*, pages 24762–24781
October 24-29, 2026 ©2026 Association for Computational Linguistics

ing knowledge, failure to use supplied evidence, and brittleness to incomplete, noisy, or misleading task inputs (Liang et al., 2023; Guha et al., 2023; Xie et al., 2024; Islam et al., 2023; Liu et al., 2024a; Yoran et al., 2024; Hong et al., 2024). These distinctions are essential because they imply different next steps: domain training (Gururangan et al., 2020), retrieval and context engineering (Lewis et al., 2020; Asai et al., 2024), verifier design (Zheng et al., 2023; Feuer et al., 2025), or targeted model adaptation. Figure 1 summarizes this diagnostic view: benchmark conditions are useful when they map model failures to different engineering decisions. This framing makes BENCHCOMPASS different from a payment-domain RAG benchmark with perturbations: the benchmark is not designed to test retrieval quality or adversarial robustness in isolation, but to decompose payment-task failures into operationally distinct signals under controlled information conditions. By pairing source-audited task construction with Closed, Open, and Attacked Open evaluation, BENCHCOMPASS turns each model–case outcome into evidence about whether the likely bottleneck is missing domain knowledge, failed use of supplied rules, or brittle rule applicability under imperfect harness inputs.

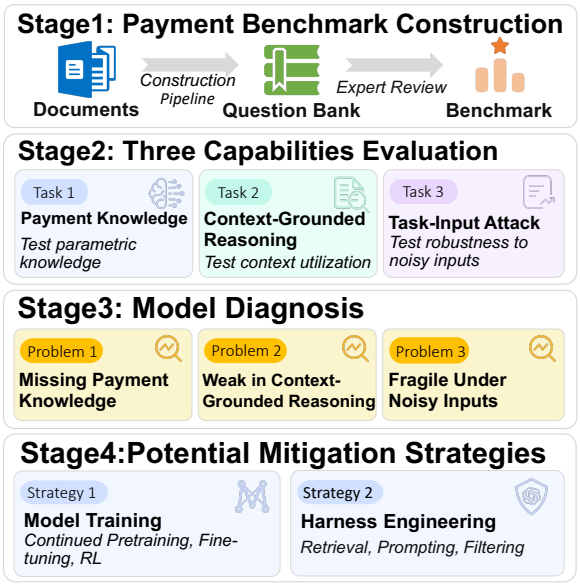



Figure 1: **Benchmark-driven diagnosis for Payment LLMs.** BENCHCOMPASS separates missing payment knowledge, failure to use supplied evidence, and brittleness to imperfect harness engineering, then maps these signals to training, retrieval, or adaptation decisions.

Constructing such a benchmark is difficult for four reasons. First, payment knowledge changes quickly, making one-off expert authoring hard to maintain. Second, valid payment-domain questions are hard to specify because the correct answer often depends on the joint state of the transaction, participant role, payment rail, jurisdiction, timing, and applicable rule hierarchy, with the supporting evidence scattered across document fragments. Third, LLMs can scale candidate construction, as shown by recent automatic benchmark-generation work (LI et al., 2025; Li et al., 2025; Bai et al., 2023), but they can also produce ambiguous, unsupported, template-like, or knowledge-free questions. Fourth, attacked-case construction and automatic scoring can create false difficulty: perturbations may change the intended answer path (Yoran et al., 2024; Hong et al., 2024; Niu et al., 2024), and LLM judges can introduce artifacts without strong quality controls (Zheng et al., 2023; Liu et al., 2023, 2024b; Gevers et al., 2025; Zheng et al., 2025; Feuer et al., 2025).

To address these challenges, we introduce BENCHCOMPASS, a domain-document-driven diagnostic benchmark construction pipeline instantiated in the payment domain. First, to reduce the maintenance burden of fast-changing payment rules, BENCHCOMPASS uses an LLM-assisted pipeline to generate source-grounded candidate cases from versioned payment documents, replacing much of the repeated manual case-authoring work while keeping experts in the admission loop. Second, to construct payment cases that are both broad and deep, BENCHCOMPASS combines typed evidence packs with 60 predefined payment scenarios and Bloom-level difficulty targets, so candidate tasks cover different payment settings while requiring controlled levels of rule understanding, application, analysis, and evaluation. Third, to control LLM generation errors, BENCHCOMPASS applies LLM-based gates for evidence support, answerability, ambiguity, difficulty, and diagnostic behavior. Fourth, to make robustness cases valid rather than merely misleading, a separate task-input attack stage creates answer-preserving attacked variants and rejects perturbations that change the intended answer path. To limit judge and construction artifacts, automatic gates only triage candidates; final Pro admission is reserved for domain experts.

Using this pipeline, we construct a domain-expert-reviewed payment Pro benchmark with 306 cases: 95 payment-knowledge MCQ, 96 Context-Grounded Reasoning open-ended QA cases for

complex payment scenarios, and 115 Task-Input Attack open-ended QA cases for robustness under answer-preserving perturbations. Our evaluation suggests that payment-domain failures are not a single capability gap. Closed/Open comparisons show that supplied rules can repair many missing-knowledge failures, but clean Open tasks remain unsaturated: the best frontier model reaches 89.6%, while a representative 32B open-weight model reaches 69.8%. On the separately constructed attacked-input set, the best frontier model reaches 81.7%, while a representative 32B open-weight model reaches 42.6%, showing that models may have the right rule in context yet still follow an invalid but plausible operational workflow. These results make BENCHCOMPASS a diagnostic benchmark: it separates when improvement requires more domain knowledge, better context-grounded reasoning, or stronger rule-applicability verification. We also retain a separately marked lower-assurance Normal pool for inspection and future curation, but all headline empirical claims use only the expert-admitted Pro benchmark.

BENCHCOMPASS makes three contributions. First, we release a domain-expert-reviewed payment-domain Pro benchmark covering payment knowledge, context-grounded scenario reasoning, and task-input Attack robustness. Second, we present an auditable document-driven construction pipeline that uses evidence packs, scenarios, and Bloom-level controls, LLM-based quality gates, task-input attacks, and final expert admission to produce auditable broad-coverage candidates before expert admission. Third, we formulate benchmark outcomes as a diagnostic model–case signals rather than aggregate scores, distinguishing failures that point to missing domain knowledge, failed use of supplied payment rules, or brittleness under imperfect harness inputs.

## 2 Related Work

**Domain benchmarks for LLMs.** General benchmarks measure broad knowledge, reasoning, and freshness (Hendrycks et al., 2021; Srivastava et al., 2023; Liang et al., 2023; Rein et al., 2023; White et al., 2025). Domain benchmarks move evaluation into specialized settings, including legal reasoning (Guha et al., 2023), broad finance (Xie et al., 2024), filing-grounded QA (Islam et al., 2023), and financial reasoning over heterogeneous evidence (Chen et al., 2021b; Zhu et al., 2021; Chen et al., 2022). These resources motivate specialized evaluation, but they do not target payment operations as a multi-party, stateful, rapidly changing infrastructure, nor do they separate payment knowledge, evidence-grounded scenario solving, and attack-input reliability.

**Automatic benchmark construction.** Recent systems use LLMs to scale benchmark construction: AutoBencher optimizes generated datasets for difficulty, novelty, and separability (LI et al., 2025); BenchBuilder curates hard prompts from interaction logs (Li et al., 2025); and Language-Model-as-an-Examiner generates and scores questions without fixed references (Bai et al., 2023). Bloom-oriented work further argues that benchmark design should control cognitive demand, not only topic coverage (Huber and Niklaus, 2025). BENCHCOMPASS differs by constructing source-grounded diagnostic cases for a changing private domain: evidence packs, scenarios, difficulty targets, and task-input attack families specify what payment knowledge, workflow condition, reasoning demand, and harness failure mode a case tests, with domain experts deciding Pro admission.

**Context use and deployment diagnosis.** Retrieval and self-reflective retrieval provide external evidence (Lewis et al., 2020; Asai et al., 2024), but models can miss, underuse, or over-rely on supplied context (Liu et al., 2024a; Bai et al., 2025; Yang et al., 2025; Goldman et al., 2024). RAG robustness studies further show that irrelevant, adversarial, conflicting, or unsupported retrieved content can degrade answers (Yoran et al., 2024; Hong et al., 2024; Park and Lee, 2024; Shen et al., 2024; Niu et al., 2024). BENCHCOMPASS is related to this literature but does not evaluate retrieval quality or claim empirical superiority over RAG-robustness benchmarks. Its evaluation target is payment-domain model behavior under fixed source availability. In particular, Attacked Open changes payment task inputs through rule-applicability modes such as sub-rule collapse, rule coverage gap, phantom state, and prior inversion, rather than simply adding distractor passages.

## 3 Task Formulation and Diagnostic Protocol

BENCHCOMPASS evaluates a model by comparing its behavior under controlled information conditions. The goal is not only to rank models, but

to separate three practical questions: whether a model already has the relevant payment knowledge, whether it can use supplied payment evidence, and whether it remains reliable when the surrounding harness provides imperfect context.

### 3.1 Evaluation Settings

We use three settings. *Closed* gives the model only the question. It tests whether the model can answer domain questions from parameters. *Open* adds sufficient context-grounded payment evidence. It evaluates whether the model can use the provided evidence to solve the task. *Attacked Open* is a separate open-book robustness setting in which the question and context may contain irrelevant, stale, conflicting, poorly ordered, or near-miss evidence. A Task-Input Attack item remains valid only when the attacked question and context still support the reference answer.

Here, "harness" refers to the input-side and checking components around a single model call: context selection, context packing and ordering, prompt framing, and rule-applicability or verifier checks. BENCHCOMPASS does not evaluate an end-to-end retriever, tool-use agent, or multi-agent decomposition pipeline; Attacked Open cases operationalize harness brittleness through answer-preserving perturbations to the question and supplied context.

### 3.2 Diagnostic Signals

Diagnostic signals are assigned to *model–case pairs*. A question is therefore not intrinsically "known" or "hard": the same case can expose different failures for different models. For base tasks, BENCHCOMPASS compares Closed and Open outcomes. For attacked-input tasks, BENCHCOMPASS reports Task-Input Attack robustness separately, because this setting tests behavior under an imperfect harness context rather than baseline knowledge access.

The repair hypotheses in Table 1 are hypotheses rather than validated causal interventions. They connect the observed signals to intervention families studied in prior work, including domain/task-adaptive training (Gururangan et al., 2020), retrieval and context engineering (Lewis et al., 2020; Asai et al., 2024), and judge or verifier design (Zheng et al., 2023; Feuer et al., 2025). Matched interventions for every diagnostic label remain future work.

## 4 Construction Pipeline

### 4.1 Pipeline Overview

Figure 2 summarizes BENCHCOMPASS. It turns a versioned payment-domain snapshot into structure-aware chunks, builds semantic profiles and evidence packs, and combines each pack with a scenario frame and Bloom difficulty level to generate base Closed/Open candidates. A base quality gate then checks ambiguity, answerability, and item quality before any attack is applied. Pattern-guided task-input attacks create Attacked Open candidates from validated answer paths, and attack-validity gates plus domain-expert admission decide whether each case enters Pro or a rejection pool.

### 4.2 Document Processing and Evidence Organization

BENCHCOMPASS represents the versioned payment-domain snapshot as structure-aware Markdown chunks with audit metadata, including source category, heading path, source location, and rule or section identifiers. The source categories cover wallet, acquirer, clearing system, card network, regulator, platform, and related payment documents; Appendix Table 5 reports the final admitted distribution. Before generation, DeepSeek V4 Pro builds a semantic profile for each chunk, summarizing its testable rule content, supported Bloom levels, and plausible payment scenarios. The pipeline then constructs evidence packs deterministically by linking each primary chunk to supporting chunks through inspectable relations such as section adjacency, shared rule identifiers, document family, payment system, currency, and profile role. Each evidence pack records the primary evidence, supporting evidence, and relation types, so reviewers can audit the source path behind every candidate task.

### 4.3 Scenario-First Base Task Generation

Base construction is scenario-first: each candidate is generated by pairing an evidence pack with an operational payment scenario and a target Bloom level. We use Bloom's revised taxonomy as the difficulty axis (Anderson and Krathwohl, 2001; Huber and Niklaus, 2025), covering L1–L5: L1 targets payment knowledge, while L2–L5 test understanding, application, analysis, and exception or precedence evaluation over supplied evidence. We exclude L6 because open-ended creation tasks are difficult to score with stable reference answers.

Table 1: **Diagnostic signals.** Base signals come from Closed/Open transitions. Task-Input Attack signals are reported separately for answer-preserving attacked items. ✓ indicates pass and ✗ indicates fail.

| Signal | Closed | Open | Interpretation | Repair Hypotheses |
|---|---|---|---|---|
| *Closed/Open base diagnosis* | | | | |
| PARAMETRIC KNOWN | ✓ | ✓ | Model parameters contain the knowledge. | Monitor drift. |
| CONTEXT RESCUED | ✗ | ✓ | Supplying context changes a Closed failure into an Open pass. | Improve retrieval/context; consider CPT/SFT. |
| CONTEXT-USE GAP | ✗ | ✗ | The model fails despite sufficient context; its context reasoning is incomplete. | Improve reasoning training, verifier, or rubric. |
| CONTEXT INTERFERENCE | ✓ | ✗ | Context hurts an otherwise correct answer. | Audit prompt, context packaging, and judge. |
| *Task-Input Attack diagnosis* | | | | |
| PERTURBATION ROBUST | – | TIA ✓ | Correct under answer-preserving perturbation. | Keep as a robustness control. |
| PERTURBATION-BRITTLE | – | TIA ✗ | Fails under answer-preserving perturbation. | Improve harness. |

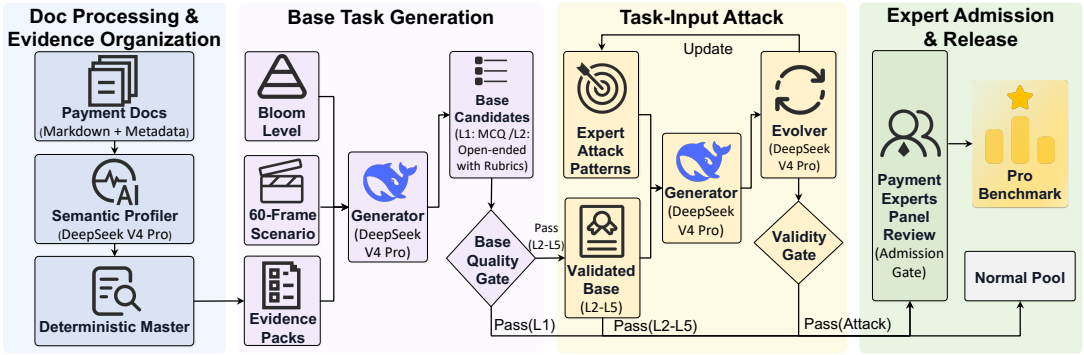



Figure 2: **BENCHCOMPASS uses scenario-first evidence-pack construction.** The system builds structure-aware chunks, semantic profiles, typed evidence packs, and grounded scenarios before generating questions.

The evidence pack anchors source support, while the scenario frame turns the source material into an operational payment problem rather than a decontextualized rule lookup. Our scenario bank contains 60 deterministic frames, with five frames for each of 12 payment-source categories, covering settings such as settlement, compliance review, disputes, product design, policy interpretation, message migration, onboarding, treasury operations, and liquidity management. DeepSeek V4 Pro then generates structured JSON candidates from the Bloom target, evidence pack, and scenario frame, with instructions to ground the question, answer, and rubric in the source text. We refer to L1 items as *Payment Knowledge* cases and L2–L5 items as *Context-Grounded Reasoning* cases. Candidates missing required answers, source quotes, choices, or rubrics are rejected. The generator uses level-specific templates rather than bare Bloom labels: L1 recall, L2 explanation, L3 rule application, L4 interacting-rule or workflow analysis, and L5 exception, conflict, or precedence resolution. Quality gates and expert admission check compliance.

### 4.4 Task-Input Attack Generation

Attacked Open tasks start from L2–L5 base items that are rescued by Open evaluation, so each seed item already has a source-supported answer path. The attack stage, therefore, tests whether a model still follows the evidence when the user-facing scenario becomes misleading or underspecified. For each seed, an attack selector chooses one or two Bloom-compatible patterns from an expert-initialized attack pool covering real-world failure scenarios; Appendix Table 8 lists both the initial and pipeline-evolved attack patterns. An LLM generator then rewrites the task input with answer-preserving perturbations, while retaining source quotes, a reference answer, and grading rubrics.

The process is iterative: validity checks and construction-time Open correctness are used only to select useful candidate variants, not as final evaluation scores. The pipeline updates attack-pattern hit rates and records newly emerged patterns across rounds. Perturbations that change the intended answer, remove the evidence path, or make the rubric ambiguous are rejected, and valid Attacked Open cases are evaluated separately from the Closed/Open diagnostic tasks.

### 4.5 Quality Control, Review, and Admission

Quality control combines general validity checks with payment-specific filters. LLM gates verify format, evidence sufficiency, answerability, ambiguity, rubric fairness, diagnostic value, and attacked-input validity. A domain-specificity gate rejects or rewrites candidates whose answer no longer depends on payment actors, states, rails, message fields, clearing or settlement concepts, or regulatory scope.

Automatic gates are used only for triage. Each finalist retains source evidence, quote maps, rubrics, model outputs, judge decisions, rejection reasons, and review metadata. Four payment-domain practitioners first jointly reviewed 30 calibration cases (10 per subset) to align a structured protocol covering question/rubric validity, judge correctness, source support, specificity, diagnostic value, duplication, and attack validity; this was protocol calibration, not full-pool IAA. The remaining candidates were partitioned into four disjoint subsets, each independently reviewed by a different practitioner, and then merged for joint adjudication. Open-book cases answered incorrectly by Gemini 3.1 Pro, GPT-5.5, or Claude Opus 4.7 received an additional two-expert validity review. Only cases with no unresolved issues were admitted; borderline, ambiguous, weakly diagnostic, duplicate, source-unstable, or defective cases were held out. The Normal pool is marked with lower assurance and retained only for inspection, development, and future curation; it is excluded from headline Pro evaluation claims.

## 5 Experiments

### 5.1 Experimental Setup

We evaluate BENCHCOMPASS on the expert-admitted Pro benchmark, covering Payment Knowledge MCQ, Context-Grounded Reasoning QA, and Task-Input Attack QA; the lower-assurance Normal pool is excluded from headline results. We test frontier closed-source, large open-weight, and mid-scale open-weight models through Alibaba Bailian and OpenRouter, using deterministic pass@1 decoding. Base tasks are evaluated in Closed and Open settings, while Task-Input Attack tasks are evaluated by attack family.

Headline OpenQA scoring uses a self-deployed DeepSeek V4 Flash judge in non-thinking mode, served on four H200 instances with four H200 GPUs each. To test judge-family sensitivity, we conduct a separate calibrated re-scoring analysis on fixed OpenQA outputs from three non-DeepSeek answer models: Gemini 3.1 Pro Preview, GLM 5.1, and Gemma 4 26B-A4B IT. DeepSeek V4 Flash, Qwen3.5-397B-A17B, and Qwen3.6-27B receive identical questions, contexts, reference answers, rubrics, and model outputs. This sensitivity analysis does not modify the headline scores. Because DeepSeek V4 Pro is used during construction and also evaluated as an answer model, we treat its score as a reference model result rather than evidence for benchmark validity. Pro admission is decided by quality gates and domain experts, and evaluated models see only the final question and source context. Full model IDs, provider routes, thinking settings, prompts, decoding settings, and judge configuration appear in Appendix D, Table 9.

Table 2: **Pro benchmark composition.** Counts are expert-admitted cases in the final benchmark.

| Subset | N | Format | Evaluation target |
|---|---|---|---|
| Payment Knowledge | 95 | MCQ | Closed/Open; L1 payment-rule knowledge |
| Context-Grounded Reasoning | 96 | Open QA | Closed/Open; L2–L5 scenarios (38/29/19/10) |
| Task-Input Attack | 115 | Open QA | Open only; answer-preserving task-input attacks |

### 5.2 Benchmark Composition

Table 2 summarizes the payment-domain release produced by BENCHCOMPASS. The final Pro benchmark contains 306 cases admitted after expert review: 95 Bloom L1 Payment Knowledge cases, 96 Bloom L2–L5 Context-Grounded Reasoning cases, and 115 Task-Input Attack cases. Payment Knowledge cases test closed-book payment knowledge. Context-Grounded Reasoning cases are evaluated in both Closed and Open settings. Task-Input Attack cases are evaluated with context and test robustness to imperfect harness engineering.

Figure 3 reports payment-domain coverage. Each case is assigned to one primary evidence class, so the counts sum to the release size. The largest categories are payment-system access and routing, regulators, currency exchange, and payment-message standards. The remaining cases cover remittance providers, banks, payment platforms, card networks, clearing and settlement, and digital wallets. This distribution reflects the source material needed for payment operations rather than a uniform topic quota. Source roles are not uniformly sampled because their operational

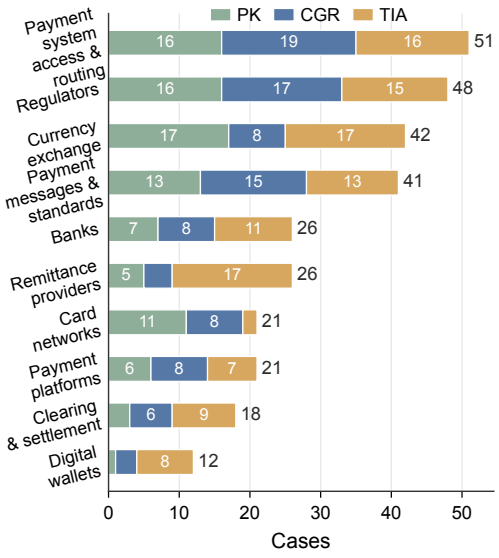



Figure 3: **Payment-domain coverage across three benchmark conditions.** Each case is counted once by its primary evidence class. PK is L1 Payment Knowledge cases, CGR is L2-L5 Context-Grounded Reasoning cases, and TIA is Task-Input Attack cases.

scope and public-document availability differ. No class dominates: Payment-system access and routing accounts for 16.7%, Regulators 15.7%, Currency exchange 13.7%, and Payment messages and standards 13.4%. We therefore treat coverage as a documented release snapshot, not an exhaustive payment-domain population. L5 contributes 10/96 CGR cases (10.4%), providing exception/precedence coverage but not supporting standalone rankings or stable level-specific conclusions. Appendix Table 6 summarizes construction statistics.

## 5.3 Main Results

Table 3 reports the main diagnostic accuracies on the expert-admitted Pro benchmark. We avoid treating small differences between nearby models as fine-grained rankings. Instead, we focus on larger patterns: Closed-to-Open gains, lower accuracy on the separately constructed Attacked Open set, and robustness gaps across model groups. Bootstrap standard errors and selected descriptive deltas are reported in Appendix Table 14. For example, the three frontier models span only 14.6–36.5% on CGR-CB but reach 82.3–89.6% on CGR-OB; these large within-condition transitions, rather than nearby leaderboard positions, are the primary empirical comparison. For these frontier models, bootstrap standard errors on CGR-CB/CGR-OB are 3.1–4.9 percentage points, whereas their Closed-to-Open gains are 45.8–75.0 points (Appendix Table 14); we therefore avoid fine-grained claims about nearby model scores.

Table 3: **Diagnostic results PK-CB**: Payment Knowledge (Closed Book); **CGR-CB**: Context-Grounded Reasoning (Closed Book); **CGR-OB**: Context-Grounded Reasoning (Open Book); **TIA-OB**: Task-Input Attack (Open Book). Models are categorized by scale and access, sorted by Task-Input Attack (OB). Because each subset contains a finite number of expert-admitted cases, we interpret small point differences cautiously. Payment Knowledge Open Book (PK-OB) is reported separately because it is saturated: 15/16 models score 95/95 (100%), and Ministral 3 8B scores 93/95 (97.9%).

| Model | PK-CB | CGR-CB | CGR-OB | TIA-OB |
|---|---|---|---|---|
| **Closed-source API Models** | | | | |
| Gemini 3.1 Pro Preview | **80.0** | 14.6 | **89.6** | **81.7** |
| GPT-5.5 | 77.9 | **36.5** | 82.3 | 80.0 |
| Claude Opus 4.7 | 68.4 | 25.0 | 82.3 | 74.8 |
| **Open-weight Large Models** | | | | |
| DeepSeek V4 Pro | 60.0 | 16.7 | 81.3 | 80.0 |
| Qwen3.5 397B-A17B | 62.1 | 14.6 | 84.4 | 72.2 |
| GLM 5.1 | 65.3 | 15.6 | 74.0 | 70.4 |
| Qwen3.5 122B-A10B | 59.0 | 9.4 | 84.4 | 70.4 |
| MiniMax M2.5 | 53.7 | 9.4 | 75.0 | 60.9 |
| Mistral Large 3 | 47.4 | 8.3 | 64.6 | 51.3 |
| Llama 4 Maverick | 54.7 | 8.3 | 68.8 | 46.1 |
| **Open-weight Compact Models** | | | | |
| Gemma 4 31B IT | 57.9 | 9.4 | 82.3 | 74.8 |
| Gemma 4 26B-A4B IT | 61.1 | 7.3 | 82.3 | 60.9 |
| Qwen3 32B | 48.4 | 5.2 | 69.8 | 42.6 |
| Ministral 3 14B | 49.5 | 2.1 | 57.3 | 37.4 |
| Qwen3 8B | 44.2 | 1.0 | 58.3 | 36.5 |
| Ministral 3 8B | 51.6 | 2.1 | 50.0 | 29.6 |

### 5.3.1 Closed/Open Diagnostic Results

The base evaluation separates accessible parametric knowledge from the ability to use the supplied payment context. The first key finding is that many payment tasks are knowledge-access failures rather than unsolvable reasoning failures. PK-CB ranges from 44.2% to 80.0%, whereas PK-OB ranges from 97.9% to 100.0%: 15 models answer all 95 items correctly, and Ministral 3 8B answers 93/95. The gap is sharper on CGR: even GPT-5.5 reaches only 36.5% in CGR-CB, and most models remain below 20%; with source context, CGR-OB rises to 50.0–89.6%. To assess wording sensitivity, we randomly audited 20 Gemini 3.1 Pro CGR-CB failures rescued in CGR-OB. In 19/20, the Closed answer omitted or misstated a substantive rule, condition, threshold, or workflow, or added an unsupported claim; only 1/20 was substantively correct but insufficiently aligned with the rubric's expected formulation. Thus, Closed-to-Open gains primarily

correct source-specific errors, although formulation sensitivity contributes occasionally. Appendix Figure 5 reports rescue rates by source class, not direct estimates of parametric-knowledge absence.

The second key finding is that Open context still leaves a real context-use gap. No model reaches 90% on CGR-OB, and the remaining errors are not mainly simple fact omissions. Across six models, manual audit identifies multi-rule synthesis and scope/exception boundaries as the leading error modes (26 model–case errors each), followed by over-expanded answer paths and modal/threshold imprecision (Appendix Table 15). Appendix Table 16 traces three cases from the missed rule to the triage implication, distinguishing context-rescued no-context failures from persistent source-grounded reasoning errors requiring better context use, verification, or adaptation.

As supplementary actionability checks on the same 95 Payment Knowledge items, a Codex corpus-access harness raised Qwen3.6-27B from 61.05% to 92.63%, while payment-domain QA SFT with 10% general data raised it to 94.74%. These checks support context access and training as actionable routes without causally validating every triage implication.

#### 5.3.2 Robustness against Task-Input Attacks

Task-Input Attack evaluation uses the fixed expert-admitted Attacked Open set. Each case remains answerable from the supplied context, but the user-facing scenario contains targeted perturbations that make the correct evidence path harder to follow. Thus, TIA-OB accuracies are final evaluation results, not construction-time attack success rates.

Table 3 reports clean CGR-OB accuracy alongside accuracy on the separately constructed TIA-OB set. Frontier models such as Gemini 3.1 Pro Preview and GPT-5.5 remain above 80% on TIA-OB, but several otherwise strong models score lower on attacked inputs than on clean Open cases. For example, Qwen3.5 397B-A17B reaches 84.4% on CGR-OB but 72.2% on TIA-OB, and Llama 4 Maverick reaches 68.8% versus 46.1%. This suggests that strong clean-context evidence use does not necessarily imply robustness to misleading task inputs, especially for lower-capacity models.

Among the better-represented modes, A-tagged cases have the lowest observed cross-model average accuracy (49.8% vs. 62.5% for B and 66.7% for E). Because labels overlap and case mixtures differ, these values are descriptive, not stable mechanism rankings. H/M/I/D_E/G (13/8/7/3/1 cases) are treated as exploratory without comparative claims. A/B/E respectively test hierarchical prerequisites, coverage boundaries, and prior inversion. Our headline robustness conclusion therefore relies on the full 115-case TIA-OB result rather than mode-level estimates.

### 5.4 Scaling and Thinking-Mode Effects

Appendix Table 13 shows that model size and thinking mode affect different diagnostic subsets differently. In the Qwen3 non-thinking sweep, L2–L5 Open accuracy rises from 53.1% at 4B to 75.0% at 235B, and Attacked Open accuracy rises from 28.7% to 51.3%, while L1 Closed remains within 50.5–58.9%. For Gemma, thinking mode improves Attacked Open accuracy for both tested models, but its effect on ordinary Open accuracy is mixed.

### 5.5 Construction Quality Analysis

Appendix Table 10 summarizes the de-duplicated expert-review finalist funnel after automatic filtering and LLM-based triage. After automated quality filtering and deduplication, 506 candidate items entered expert review; four payment-domain experts admitted 306 of them to Pro (60.5%). Appendix Table 11 summarizes the primary reasons for holding out the remaining candidates. To assess admission reliability, 60 randomly sampled candidates (20 per subset) were blindly re-reviewed under the same criteria by an original practitioner who had not conducted that item's first-pass review. Binary agreement was 57/60 (95.0%; Cohen's $\kappa = 0.900$); the three disagreements involved borderline provider specificity or diagnostic value, not unsupported answers or incorrect rubrics. To check generator portability, we replaced DeepSeek V4 Pro with Qwen3.5-397B-A17B while keeping the downstream pipeline unchanged. Of 100 generated candidates, 62 were usable after downstream QC and conservative manual review. This shows that the code path can produce usable candidates with another strong model, but does not establish generator-model invariance.

### 5.6 Judging Reliability Checks

Four human reviewers audited 255 visible-verdict OpenQA judge decisions: all 115 Attacked Open decisions and 70 sampled decisions from each of Base Closed and Base Open. For open-book cases failed by frontier models, two reviewers cross-checked the case and judge decision; disagreement

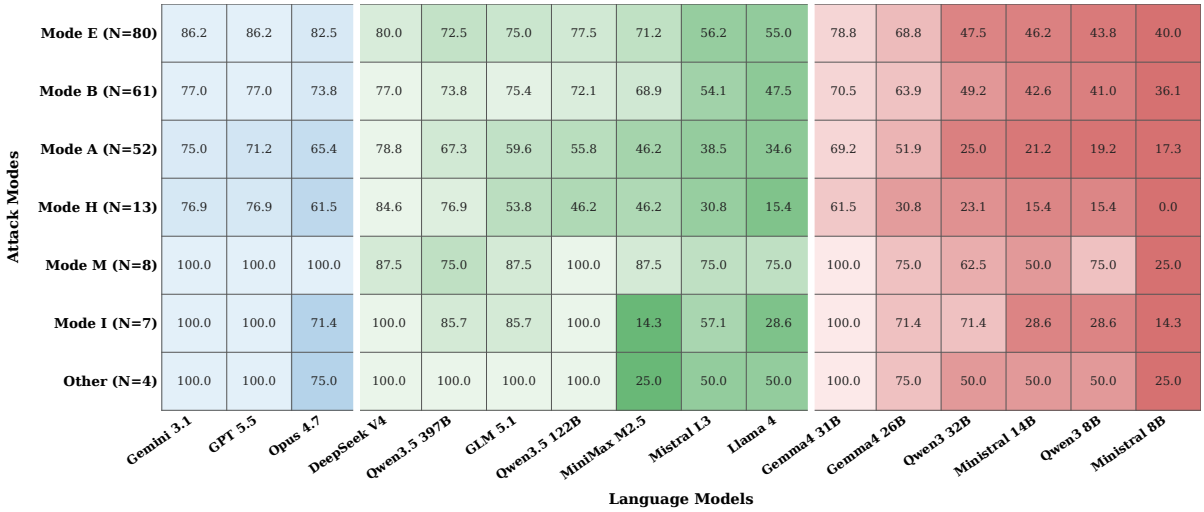



Figure 4: **Robustness by attack mode.** Cells report descriptive accuracy (%) for cases carrying each attack-mode label. Labels overlap: 110 of the 115 cases carry two labels, and five carry one, yielding 225 mode mentions. A/B/E are the better-represented modes; H/M/I/D_E/G(Other) are low-count and exploratory. Darker cells indicate lower accuracy. Attack definitions are provided in Appendix Table 8.

Table 4: **Human audit of OpenQA judging.** Accept agrees with the judge, Partial marks a minor issue, and Wrong marks an incorrect judge decision; MCQ items are excluded.

| Evaluation | N | Accept | Partial | Wrong | Accept rate |
|---|---|---|---|---|---|
| Base Closed | 70 | 66 | 3 | 1 | 94.3 |
| Base Open | 70 | 70 | 0 | 0 | 100.0 |
| Attacked Open | 115 | 115 | 0 | 0 | 100.0 |
| **All OpenQA** | **255** | **251** | **3** | **1** | **98.4** |

made the case ineligible for the benchmark. As shown in Table 4, reviewers accepted 251/255 decisions (98.4%); all audited Base Open and Attacked Open decisions were accepted.

Under a shared calibrated protocol, Qwen3.5-397B and Qwen3.6-27B re-score 921 fixed OpenQA outputs from Gemini 3.1 Pro, GLM 5.1, and Gemma 4 26B-A4B. Their agreement with DeepSeek V4 Flash is 93.05% and 91.97%, with correctness-rate shifts of $-2.17$ and $-1.74$ points; per-model agreement exceeds 91% (Appendix Table 12). This separate sensitivity analysis reduces judge-family concerns without replacing headline scores and cautions against overinterpreting small differences.

For same-judge repeatability, three DeepSeek V4 Flash re-scorings of fixed GPT-5.5, GLM 5.1, and Qwen3 32B outputs across CGR-CB/CGR-OB/TIA-OB produce parseable JSON for all 2,763 calls and a maximum accuracy range of 2.08 points, well below the main Closed/Open transitions.

## 6 Conclusion

We introduced BENCHCOMPASS, a payment-domain benchmark and construction pipeline for evaluating LLMs on dynamic, document-grounded payment workflows. The benchmark shows that payment failures are not a single capability gap: models may lack domain knowledge, fail to compose supplied rules, or follow a plausible but invalid workflow despite having the relevant context. By separating these behaviors through Closed, Open, and Attacked Open settings, BENCHCOMPASS turns benchmark scores into diagnostic signals that can help prioritize future work on domain training, context use, and rule-applicability verification.

## Limitations

This paper studies payment-domain workflows rather than general finance. The Pro benchmark is expert-reviewed, but it is still a finite snapshot of public or release-reviewable payment documents, so scores should be interpreted relative to the released source snapshot rather than as permanent model properties.

BENCHCOMPASS evaluates use of supplied context, not end-to-end retrieval. The Closed, Open, and Attacked Open labels are behavioral diagnostic signals rather than causal explanations; matched repair experiments are needed to verify which interventions improve specific failure sets. Bloom levels and attack patterns are coverage devices, not a com-

plete theory of payment reasoning. OpenQA scoring uses structured LLM judging with human audit and a secondary-judge check, but residual rubric sensitivity, judging errors, and generator/judge model-family overlap may remain.

## Ethical Considerations

BENCHCOMPASS is intended for benchmark evaluation and engineering diagnosis, not for payment, financial, legal, compliance, or customer-facing advice. The benchmark uses public or release-reviewed payment documents and does not use private customer records, transaction logs, account data, or personally identifiable information. Cases mentioning real institutions test the interpretation of public rules, not institutional behavior or risk.

The release includes runnable benchmark cases, evidence snippets, source provenance, reference answers, rubrics, prompts, model outputs, judge decisions, human audit labels, and run configuration. Full original documents are redistributed only when licensing permits. Attacked Open cases are answer-preserving robustness probes, not instructions for fraud, evasion, or bypassing controls. Human reviewers were informed of research use, reviewer identities are not released, and AI-assisted construction, coding, analysis, and editing were reviewed by the authors before release.

## References


Lorin W Anderson and David R Krathwohl. 2001. *A taxonomy for learning, teaching, and assessing: A revision of Bloom's taxonomy of educational objectives: complete edition*. Addison Wesley Longman, Inc.

Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. 2024. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. In *International Conference on Learning Representations*.

Yushi Bai, Shangqing Tu, Jiajie Zhang, Hao Peng, Xiaozhi Wang, Xin Lv, Shulin Cao, Jiazheng Xu, Lei Hou, Yuxiao Dong, Jie Tang, and Juanzi Li. 2025. LongBench v2: Towards deeper understanding and reasoning on realistic long-context multitasks. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3639–3664, Vienna, Austria. Association for Computational Linguistics.

Yushi Bai, Jiahao Ying, Yixin Cao, Xin Lv, Yuze He, Xiaozhi Wang, Jifan Yu, Kaisheng Zeng, Yijia Xiao, Haozhe Lyu, Jiayin Zhang, Juanzi Li, and Lei Hou. 2023. Benchmarking foundation models with language-model-as-an-examiner. In *Advances in Neural Information Processing Systems*.

Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, Alex Ray, Raul Puri, Gretchen Krueger, Michael Petrov, Heidy Khlaaf, Girish Sastry, Pamela Mishkin, Brooke Chan, Scott Gray, Nick Ryder, Mikhail Pavlov, Alethea Power, Lukasz Kaiser, Mohammad Bavarian, Clemens Winter, Philippe Tillet, Felipe Petroski Such, Dave Cummings, Matthias Plappert, Fotios Chantzis, Elizabeth Barnes, Ariel Herbert-Voss, William Hebgen Guss, Alex Nichol, Alex Paino, Nikolas Tezak, Jie Tang, Igor Babuschkin, Suchir Balaji, Shantanu Jain, William Saunders, Christopher Hesse, Andrew N. Carr, Jan Leike, Josh Achiam, Vedant Misra, Evan Morikawa, Alec Radford, Matthew Knight, Miles Brundage, Mira Murati, Katie Mayer, Peter Welinder, Bob McGrew, Dario Amodei, Sam McCandlish, Ilya Sutskever, and Wojciech Zaremba. 2021a. Evaluating large language models trained on code. *arXiv preprint arXiv:2107.03374*.

Zhiyu Chen, Wenhu Chen, Charese Smiley, Sameena Shah, Iana Borova, Dylan Langdon, Reema Moussa, Matt Beane, Ting-Hao Huang, Bryan Routledge, and William Yang Wang. 2021b. FinQA: A dataset of numerical reasoning over financial data. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 3697–3711, Online and Punta Cana, Dominican Republic. Association for Computational Linguistics.

Zhiyu Chen, Shiyang Li, Charese Smiley, Zhiqiang Ma, Sameena Shah, and William Yang Wang. 2022. ConvFinQA: Exploring the chain of numerical reasoning in conversational finance question answering. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 6279–6292, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.

Benjamin Feuer, Chiung-Yi Tseng, Astitwa Sarthak Lathe, Oussama Elachqar, and John P. Dickerson. 2025. When judgment becomes noise: How design failures in LLM judge benchmarks silently undermine validity. *arXiv preprint arXiv:2509.20293*.

Ine Gevers, Victor De Marez, Jens Van Nooten, Jens Lemmens, Andriy Kosar, Ehsan Lotfi, Nikolay Banar, Pieter Fivez, Luna De Bruyne, and Walter Daelemans. 2025. In benchmarks we trust ... or not? In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 23662–23676, Suzhou, China. Association for Computational Linguistics.

Omer Goldman, Alon Jacovi, Aviv Slobodkin, Aviya Maimon, Ido Dagan, and Reut Tsarfaty. 2024. Is it really long context if all you need is retrieval? towards genuinely difficult long context NLP. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 16576–16586,

Miami, Florida, USA. Association for Computational Linguistics.

Neel Guha, Julian Nyarko, Daniel E. Ho, Christopher Ré, Adam S. Chilton, Alex Chohlas-Wood, Austin Peters, Brandon Waldon, Daniel Rockmore, Diego Zambrano, et al. 2023. LegalBench: A collaboratively built benchmark for measuring legal reasoning in large language models. In *Advances in Neural Information Processing Systems*.

Suchin Gururangan, Ana Marasović, Swabha Swayamdipta, Kyle Lo, Iz Beltagy, Doug Downey, and Noah A. Smith. 2020. Don't stop pretraining: Adapt language models to domains and tasks. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 8342–8360, Online. Association for Computational Linguistics.

Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. 2021. Measuring massive multitask language understanding. In *International Conference on Learning Representations*.

Giwon Hong, Jeonghwan Kim, Junmo Kang, Sung-Hyon Myaeng, and Joyce Jiyoung Whang. 2024. Why so gullible? enhancing the robustness of retrieval-augmented models against counterfactual noise. In *Findings of the Association for Computational Linguistics: NAACL 2024*, pages 2474–2495, Mexico City, Mexico. Association for Computational Linguistics.

Thomas Huber and Christina Niklaus. 2025. LLMs meet bloom's taxonomy: A cognitive view on large language model evaluations. In *Proceedings of the 31st International Conference on Computational Linguistics*, pages 5211–5246, Abu Dhabi, UAE. Association for Computational Linguistics.

Pranab Islam, Anand Kannappan, Douwe Kiela, Rebecca Qian, Nino Scherrer, and Bertie Vidgen. 2023. FinanceBench: A new benchmark for financial question answering. *arXiv preprint arXiv:2311.11944*.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*.

Tianle Li, Wei-Lin Chiang, Evan Frick, Lisa Dunlap, Tianhao Wu, Banghua Zhu, Joseph E. Gonzalez, and Ion Stoica. 2025. From crowdsourced data to high-quality benchmarks: Arena-Hard and BenchBuilder pipeline. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 34209–34231. PMLR.

XIANG LI, Farzaan Kaiyom, Evan Liu, Yifan Mai, Percy Liang, and Tatsunori Hashimoto. 2025. Autobencher: Towards declarative benchmark construction. In *International Conference on Learning Representations*, volume 2025, pages 94357–94386.

Percy Liang, Rishi Bommasani, Tony Lee, Dimitris Tsipras, Dilara Soylu, Michihiro Yasunaga, Yian Zhang, Deepak Narayanan, Yuhuai Wu, Ananya Kumar, et al. 2023. Holistic evaluation of language models. *Transactions on Machine Learning Research*.

Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2024a. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173.

Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. 2023. G-eval: NLG evaluation using gpt-4 with better human alignment. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 2511–2522, Singapore. Association for Computational Linguistics.

Yu Lu Liu, Su Lin Blodgett, Jackie Cheung, Q. Vera Liao, Alexandra Olteanu, and Ziang Xiao. 2024b. ECBD: Evidence-centered benchmark design for NLP. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 16349–16365, Bangkok, Thailand. Association for Computational Linguistics.

Cheng Niu, Yuanhao Wu, Juno Zhu, Siliang Xu, KaShun Shum, Randy Zhong, Juntong Song, and Tong Zhang. 2024. RAGTruth: A hallucination corpus for developing trustworthy retrieval-augmented language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 10862–10878, Bangkok, Thailand. Association for Computational Linguistics.

Seong-Il Park and Jay-Yoon Lee. 2024. Toward robust RALMs: Revealing the impact of imperfect retrieval on retrieval-augmented language models. *Transactions of the Association for Computational Linguistics*, 12:1686–1702.

David Rein, Betty Li Hou, Asa Cooper Stickland, Jackson Petty, Richard Yuanzhe Pang, Julien Dirani, Julian Michael, and Samuel R. Bowman. 2023. GPQA: A graduate-level google-proof Q&A benchmark. *arXiv preprint arXiv:2311.12022*.

Xiaoyu Shen, Rexhina Blloshmi, Dawei Zhu, Jiahuan Pei, and Wei Zhang. 2024. Assessing "implicit" retrieval robustness of large language models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 8988–9003, Miami, Florida, USA. Association for Computational Linguistics.

Aarohi Srivastava, Abhinav Rastogi, Abhishek Rao, Abu Awal Md Shoeb, Abubakar Abid, Adam Fisch, Adam R. Brown, Adam Santoro, Aditya Gupta, Adria Garriga-Alonso, et al. 2023. Beyond the imitation game: Quantifying and extrapolating the capabilities of language models. *Transactions on Machine Learning Research*.

Colin White, Samuel Dooley, Manley Roberts, Arka Pal, Benjamin Feuer, Siddhartha Jain, Ravid Shwartz-Ziv, Neel Jain, Khalid Saifullah, Sreemanti Dey, et al. 2025. Livebench: A challenging, contamination-limited llm benchmark. In *International Conference on Learning Representations*, volume 2025, pages 91595–91631.

Qianqian Xie, Weiguang Han, Zhengyu Chen, Ruoyu Xiang, Xiao Zhang, Yueru He, Mengxi Xiao, Dong Li, Yongfu Dai, Duanyu Feng, et al. 2024. FinBen: A holistic financial benchmark for large language models. In *Advances in Neural Information Processing Systems*.

Van Yang, Hongye Jin, Shaochen Zhong, Song Jiang, Qifan Wang, Vipin Chaudhary, and Xiaotian Han. 2025. 100-LongBench: Are de facto long-context benchmarks literally evaluating long-context ability? In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 17560–17576, Vienna, Austria. Association for Computational Linguistics.

Ori Yoran, Tomer Wolfson, Ori Ram, and Jonathan Berant. 2024. Making retrieval-augmented language models robust to irrelevant context. In *International Conference on Learning Representations*.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, et al. 2023. Judging llm-as-a-judge with mt-bench and chatbot arena. *Advances in neural information processing systems*, 36:46595–46623.

Xiaosen Zheng, Tianyu Pang, Chao Du, Qian Liu, Jing Jiang, and Min Lin. 2025. Cheating automatic llm benchmarks: Null models achieve high win rates. In *International Conference on Learning Representations*, volume 2025, pages 61994–62027.

Fengbin Zhu, Wenqiang Lei, Youcheng Huang, Chao Wang, Shuo Zhang, Jiancheng Lv, Fuli Feng, and Tat-Seng Chua. 2021. TAT-QA: A question answering benchmark on a hybrid of tabular and textual content in finance. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 3277–3287, Online. Association for Computational Linguistics.

# A Payment Source Coverage

Table 5: **Payment source categories represented in the Pro benchmark.** Categories match Figure 3. Evidence documents are distinct source files within each category; the total row reports unique source files across the admitted Pro benchmark.

| Source category | Ev. docs | L1 | L2–L5 | Attack | Total | Typical coverage |
|---|---|---|---|---|---|---|
| Payment system access & routing | 11 | 16 | 19 | 16 | 51 | Participant access, routing, FPS/CHATS/RTGS-like systems, payment rails, and operational flow. |
| Regulators | 11 | 16 | 17 | 15 | 48 | Licensing, AML/CFT, consumer protection, data protection, and supervisory rules. |
| Currency exchange | 11 | 17 | 8 | 17 | 42 | FX conversion, exchange-rate disclosure, transfer limits, and cross-border currency constraints. |
| Payment messages & standards | 15 | 13 | 15 | 13 | 41 | ISO 20022, SWIFT-like message standards, field requirements, and technical guidance. |
| Banks | 11 | 7 | 8 | 11 | 26 | Commercial and digital bank account terms, e-banking, onboarding, and account-state rules. |
| Remittance providers | 13 | 5 | 4 | 17 | 26 | Transfer flow, fees, delivery channels, arrival time, refunds, and service restrictions. |
| Card networks | 4 | 11 | 8 | 2 | 21 | Card-network rules, dispute handling, chargeback, tokenization, and merchant constraints. |
| Payment platforms | 6 | 6 | 8 | 7 | 21 | Cross-border collection, merchant settlement, PSP accounts, notifications, and platform risk controls. |
| Clearing & settlement | 6 | 3 | 6 | 9 | 18 | Clearing-house rules, settlement timing, value date, reconciliation, and exception handling. |
| Digital wallets | 6 | 1 | 3 | 8 | 12 | Stored value, top-up, withdrawal, QR payment, KYC, and wallet transfer rules. |
| **Total** | **73** | **95** | **96** | **115** | **306** | **Final domain-expert-reviewed Pro benchmark.** |

Table 6: **Benchmark statistics and prompt targets.** Context is measured in characters. Source segments count the flattened 'Source' blocks in the released context and summarize evidence-pack size. Rubric counts are means for OpenQA.

| Subset | N | Bloom distribution | Source seg. | Context chars | Rubrics | Prompt target |
|---|---|---|---|---|---|---|
| Payment Knowledge | 95 | L1: 95 | 1.0 | 1,180 | – | Recall or recognize a source-specific payment rule in a self-contained four-choice MCQ. |
| Context-Grounded Reasoning | 96 | L2/L3/L4/L5: 38/29/19/10 | 1.9 | 3,116 | 3.1 | Explain, apply, analyze, or resolve exceptions/precedence over a supplied evidence pack. |
| Task-Input Attack | 115 | L2/L3/L4/L5: 18/88/6/3 | 1.9 | 3,593 | 3.2 | Apply one or two answer-preserving rule-applicability perturbations to a validated seed. |

## B Case Metadata Schema

Table 7: **Release metadata schema.** Required fields are included in the public benchmark package. Audit fields are released to support reproducibility and error analysis when they do not expose restricted source material or reviewer-identifying information.

| Field group | Status | Fields |
|---|---|---|
| Identity and split | Required | Case ID, original construction ID when available, release split, benchmark subset, question type, Bloom level, and diagnostic category. |
| Evaluation input | Required | User-facing question or task input, source-grounded context for Open or Attacked Open cases, and choices for multiple-choice cases. |
| Answer and scoring | Required | Reference answer, grading rubrics for OpenQA, attack pattern codes for Attacked Open cases, and deterministic parsing rules for MCQ cases. |
| Source provenance and evidence | Required | Source-grounded evidence snippets used for evaluation, source category, source document ID, public URL when available, retrieval date, and evidence identifiers; full original documents are redistributed only when licensing permits. |
| Construction metadata | Audit release | Evidence-pack roles, scenario frame, chunk profile summaries, generation prompts, quality-gate outputs, and rejection reasons, with restricted source text or reviewer-identifying notes removed when needed. |
| Evaluation artifacts | Audit release | Evaluation prompts, model outputs, final judge decisions, manual audit labels, review notes when releasable, and run configuration; raw chain-of-thought is not stored or released. |

## C Detailed Attack Patterns

Table 8: **Detailed Attack Patterns for Task-Input Attack Item Generation.** The table lists both expert-initialized and pipeline-evolved attack modes, detailing their disruptive mechanisms and construction logic. Mode E denotes Prior Inversion, whereas D_E denotes Semantic Split + Prior Inversion; Figure 4 combines only the low-count D_E/G display rows as D_E/G (3/1).

| Mode | Name | Mechanism | Construction | Source |
|---|---|---|---|---|
| A | Sub-rule Premise Collapse | Global sub-rule S2 presupposes that parent rule S1 is universally valid. When the domain of S1 is narrowed, S2's premise vanishes, but the model still mechanically triggers S2. | (1) Find a rule where a subset prioritizes a specific action; (2) narrow the eligible pool to a single type so that S1 vanishes; (3) retain triggers for S2; (4) correct behavior: fallback to general sorting; error: applies S2 mechanically. | Domain Experts |
| B | Rule Coverage Gap Over-generalization | Rule R covers general cases but remains silent on the current edge case E. The model logically extends R to fill the silence without realizing it is inventing a rule. | (1) Find rule R lacking precise boundary constraints; (2) place a distant rule $R'$ implicitly governing E; (3) design scenario E to fall inside R's silent zone while appearing to be a general case. | Domain Experts |
| C | /Sub-section Override | The main table provides base definitions, and the model stops reading once processed. Critical constraints are hidden in distant sub-sections. | Choose values that trigger the hidden sub-section constraints. Ensure the sub-section is physically separated from the main table by other content. | Domain Experts |
| D_E | Semantic Split + Prior Inversion | Two segments share identical topic structures but differ in one qualifier, yielding opposite conclusions. The correct conclusion contradicts common sense. | (1) Find rules R-alpha and R-beta with identical topics but opposite conclusions; (2) design a scenario triggering R-beta; (3) make R-alpha's wording prominent. | Domain Experts |
| E | Prior Inversion | A knowledge-base rule conflicts with pre-trained common-sense expectations. The model finds the rule but silently overrides it. | (1) Find a rule contradicting common sense; (2) design a scenario where prior expectations conflict with the rule; (3) emphasize "according to the rule" in the prompt. | Domain Experts |
| F | Symmetry Bias | The correct answer requires treating superficially identical objects differently. The model tends to give a symmetric "both can/cannot" response. | Design a situation where A and B belong to the same category, but A can perform an action while B cannot. The correct answer must treat them asymmetrically. | Domain Experts |
| G | Total vs. Batch / Sub-account | A rule is triggered by insufficient sub-account or batch balances, but the model incorrectly uses the total balance for its judgment. | Initialize an insufficient sub-account balance but sufficient total balance. Emphasize "balance" rather than "batch balance" in the prompt to induce confusion. | Pipeline Evolution |
| H | Phantom State Mutation | An operation triggers an implicit global hook to switch the system into state $S'$. The model ignores $S'$ and processes subsequent steps using the original state. | (1) Find an event switching the system to state $S'$; (2) hide this event in intermediate steps; (3) ask questions depending on state $S'$ rules, while hiding the trigger in sub-sections. | Pipeline Evolution |
| I | Causal Graph Hijacking | The model has a strong causal prior $A \rightarrow C$. System rules insert block B. The scenario bypasses B, meaning A cannot reach C. | (1) Identify a strong causal prior; (2) introduce M where X leads to Y only if M is passed; (3) design a scenario bypassing M via obscure rules, thereby exploiting the model's faulty prior. | Pipeline Evolution |
| M | Polysemic Drift | Identical business terms are assigned different precise meanings across documents. The model incorrectly mixes these contextual meanings. | Find identical terms with varying definitions (e.g., the calculation of "application date"). Construct a scenario at the boundary that requires contextual word-sense resolution. | Pipeline Evolution |

# D Evaluation Model Settings

Table 9: **Model invocation settings for the final merged evaluation.** All answer-model calls use pass@1 and temperature 0. The final report keeps provider-default, explicit non-thinking, and supplemental thinking-enabled runs separate when multiple modes were evaluated.

| Reported model or group | Platform | API model ID(s) | Mode in final report | Reasoning / storage note |
|---|---|---|---|---|
| GPT-5.5 | OpenRouter | `openai/gpt-5.5` | default/thinking enabled | Provider usage reports hidden reasoning tokens for most rows. |
| Gemini 3.1 Pro Preview | OpenRouter | `google/gemini-3.1-pro-preview` | default/thinking enabled | Provider usage reports hidden reasoning tokens for a subset of rows. |
| Claude Opus 4.7 | OpenRouter | `anthropic/claude-opus-4.7` | thinking enabled | Enables reasoning; no raw chain-of-thought is stored. |
| DeepSeek V4 Pro | Bailian | `deepseek-v4-pro` | default | Bailian default setting was used; older merged rows do not expose reasoning-token metadata. |
| Qwen3.5 family | Bailian | Qwen3.5 397B-A17B, 122B-A10B | default | Provider-default hybrid-thinking mode; usage metadata reports reasoning tokens when available. |
| GLM 5.1 | Bailian | `glm-5.1` | default | Provider-default mode; usage metadata reports reasoning tokens for a subset of rows. |
| MiniMax M2.5 | Bailian | `MiniMax-M2.5` | default | Provider-default mode; usage metadata reports reasoning tokens for a subset of rows. |
| Qwen3 open-weight scale sweep | Bailian | Qwen3 4B, 8B, 14B, 32B, 235B-A22B | non-thinking; thinking enabled | Non-thinking rows use `enable_thinking=false`; supplemental rows use explicit thinking. Both are reported for scale analysis. |
| Gemma 4 IT | OpenRouter | Gemma 4 31B IT; 26B-A4B IT | default / no explicit thinking; thinking enabled | Both rows are reported. Supplemental rows explicitly enable reasoning through OpenRouter. |
| Other overseas open models | OpenRouter | Llama 4 Maverick; Mistral Large 3; Ministral 3 14B/8B | default | No explicit reasoning parameter was sent; no reasoning-token metadata was observed for these exact model IDs. |
| Judge | self-deployed | DeepSeek V4 Flash | final structured verdict only | OpenQA answers are judged by DeepSeek V4 Flash. MCQ uses deterministic option parsing. The judge stores final JSON decisions only, not reasoning traces. |
| Cross-family judge sensitivity | Self-deployed | Qwen3.5 397B-A17B; Qwen3.6-27B | non-thinking; temperature 0 | Calibrated rubric-first re-scoring of fixed outputs from three non-DeepSeek answer models; not used for headline scoring. |

# E Additional Experiment Results

Table 10: **Construction funnel and expert admission.** Counts are de-duplicated expert-review finalists, not raw iterative generation attempts. The Normal pool is lower assurance and excluded from Pro evaluation.

| Stage | Finalists | Pro | Held out | Admit. % | Normal |
|---|---|---|---|---|---|
| Base | 356 | 191 | 165 | 53.7 | 891 |
| Attack | 150 | 115 | 35 | 76.7 | 355 |
| Total | 506 | 306 | 200 | 60.5 | 1,246 |

Table 11: **Primary exclusion reasons for the 200 held-out finalists.** Each item is assigned one primary reason from expert labels and written comments.

| Primary reason | Share |
|---|---|
| Provider-specific or low-transferability rule | 42.5% |
| Volatile or under-supported source grounding | 14.5% |
| Answer–rubric mismatch or overly strict grading | 13.0% |
| Ambiguous or underspecified task conditions | 12.5% |
| Near-duplicate or coverage-balancing hold-out | 7.0% |
| Business-realism or sensitivity concern | 5.5% |
| Other borderline issue | 3.0% |
| Weak diagnostic value | 2.0% |

Table 12: **Cross-family judge sensitivity on fixed OpenQA outputs.** All three judges are applied under the same calibrated rubric-first re-scoring protocol; this analysis does not alter headline scores. Each Qwen cell reports correct outputs, score shift relative to DeepSeek V4 Flash, and item-level agreement.

| Answer model | N | DeepSeek correct | Qwen3.5-397B | Qwen3.6-27B |
|---|---|---|---|---|
| GLM 5.1 | 307 | 205/307 | 201/307; $-1.30$ pp; $94.14\%$ | 206/307; $+0.33$ pp; $91.21\%$ |
| Gemma 4 26B-A4B IT | 307 | 195/307 | 186/307; $-2.93$ pp; $93.81\%$ | 181/307; $-4.56$ pp; $93.49\%$ |
| Gemini 3.1 Pro Preview | 307 | 225/307 | 218/307; $-2.28$ pp; $91.21\%$ | 222/307; $-0.98$ pp; $91.21\%$ |
| **Pooled** | **921** | **625/921** | **605/921;** $-2.17$ **pp;** $93.05\%$ | **609/921;** $-1.74$ **pp;** $91.97\%$ |

Table 13: **Supplemental thinking-mode results.** Values are accuracies on the final Pro benchmark. L1-C is L1 Closed accuracy; L2-O is L2–L5 Open accuracy; Attack-O is Attacked Open accuracy.

| Model | Mode | L1-C | L2-O | Attack-O | Note |
|---|---|---|---|---|---|
| *Qwen3 thinking-mode size sweep* | | | | | |
| Qwen3 4B | thinking | 43.2 | 51.0 | 26.1 | Smallest model remains weakest under attacked inputs. |
| Qwen3 8B | thinking | 44.2 | 58.3 | 36.5 | Open evidence use improves, but robustness remains limited. |
| Qwen3 14B | thinking | 43.2 | 58.3 | 46.1 | Better attacked-input robustness than 8B. |
| Qwen3 32B | thinking | 48.4 | 69.8 | 42.6 | Highest Open accuracy in the Qwen3 thinking sweep. |
| Qwen3 235B-A22B | thinking | 55.8 | 61.5 | 47.0 | Best attacked-input accuracy in the Qwen3 thinking sweep. |
| *Other supplemental thinking-mode runs* | | | | | |
| Claude Opus 4.7 | thinking | 68.4 | 82.3 | 74.8 | Strong attacked-input robustness among supplemental runs. |
| Gemma 4 31B IT | thinking | 57.9 | 82.3 | 74.8 | Matches Claude Opus 4.7 on Open and Attack-O in this run. |
| Gemma 4 26B-A4B IT | thinking | 61.1 | 82.3 | 60.9 | Strong Open accuracy with weaker attacked-input robustness. |
| Kimi K2.6 | thinking | 66.3 | 90.6 | 73.0 | Highest L2–L5 Open accuracy in the supplemental thinking runs. |

Table 14: **Bootstrap uncertainty estimates.** Values are percentages, and SE denotes bootstrap standard error over benchmark cases. $\Delta$ is CGR-OB minus TIA-OB; because these are separate subsets, its SE is a descriptive independent-resampling estimate.

| Model | PK-CB | SE | CGR-CB | SE | CGR-OB | SE | TIA-OB | SE | $\Delta$ | SE |
|---|---|---|---|---|---|---|---|---|---|---|
| Gemini 3.1 Pro Preview | 80.0 | 4.1 | 14.6 | 3.6 | 89.6 | 3.1 | 81.7 | 3.6 | 7.9 | 4.8 |
| GPT-5.5 | 77.9 | 4.3 | 36.5 | 4.9 | 82.3 | 3.9 | 80.0 | 3.7 | 2.3 | 5.4 |
| Claude Opus 4.7 | 68.4 | 4.8 | 25.0 | 4.4 | 82.3 | 3.9 | 74.8 | 4.0 | 7.5 | 5.6 |
| DeepSeek V4 Pro | 60.0 | 5.0 | 16.7 | 3.8 | 81.3 | 4.0 | 80.0 | 3.7 | 1.3 | 5.5 |
| Qwen3.5 397B-A17B | 62.1 | 5.0 | 14.6 | 3.6 | 84.4 | 3.7 | 72.2 | 4.2 | 12.2 | 5.6 |
| GLM 5.1 | 65.3 | 4.9 | 15.6 | 3.7 | 74.0 | 4.5 | 70.4 | 4.3 | 3.6 | 6.2 |
| Qwen3.5 122B-A10B | 59.0 | 5.0 | 9.4 | 3.0 | 84.4 | 3.7 | 70.4 | 4.3 | 14.0 | 5.6 |
| MiniMax M2.5 | 53.7 | 5.1 | 9.4 | 3.0 | 75.0 | 4.4 | 60.9 | 4.6 | 14.1 | 6.3 |
| Mistral Large 3 | 47.4 | 5.1 | 8.3 | 2.8 | 64.6 | 4.9 | 51.3 | 4.7 | 13.3 | 6.7 |
| Llama 4 Maverick | 54.7 | 5.1 | 8.3 | 2.8 | 68.8 | 4.7 | 46.1 | 4.6 | 22.7 | 6.6 |
| Gemma 4 31B IT | 57.9 | 5.1 | 9.4 | 3.0 | 82.3 | 3.9 | 74.8 | 4.0 | 7.5 | 5.6 |
| Gemma 4 26B-A4B IT | 61.1 | 5.0 | 7.3 | 2.7 | 82.3 | 3.9 | 60.9 | 4.6 | 21.4 | 6.0 |
| Qwen3 32B | 48.4 | 5.1 | 5.2 | 2.3 | 69.8 | 4.7 | 42.6 | 4.6 | 27.2 | 6.6 |
| Ministral 3 14B | 49.5 | 5.1 | 2.1 | 1.5 | 57.3 | 5.0 | 37.4 | 4.5 | 19.9 | 6.8 |
| Qwen3 8B | 44.2 | 5.1 | 1.0 | 1.0 | 58.3 | 5.0 | 36.5 | 4.5 | 21.8 | 6.7 |
| Ministral 3 8B | 51.6 | 5.1 | 2.1 | 1.5 | 50.0 | 5.1 | 29.6 | 4.3 | 20.4 | 6.6 |

# F Closed/Open Diagnostic Details

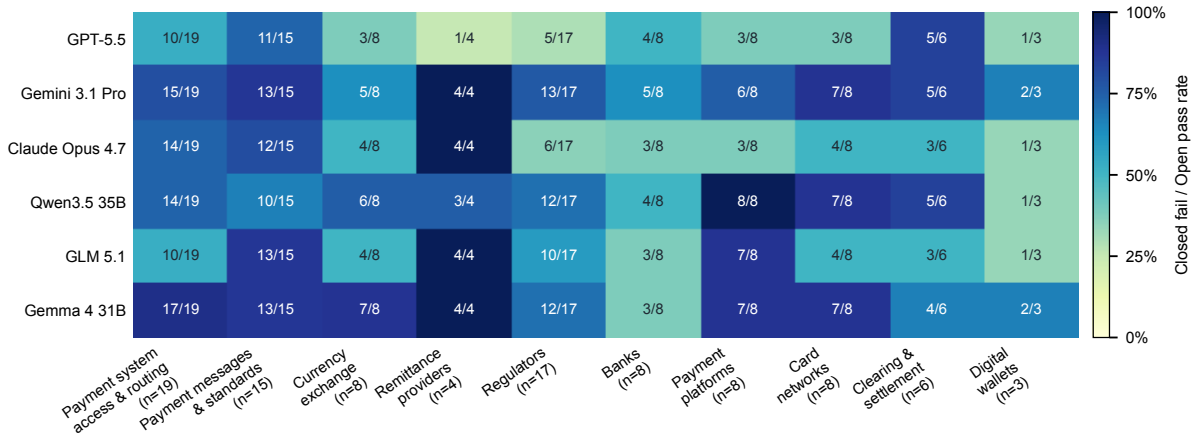



Figure 5: **Where Open context rescues representative models.** Each cell counts L2–L5 cases where a model fails Closed but passes Open, grouped by the case's primary source category. Larger counts indicate source areas where the model lacks accessible parametric knowledge but can use the supplied source context.

Table 15: **Manual audit of representative Open failures.** Counts are model–case errors for six representative models on L2–L5 Open cases. The audit separates substantive context-use errors from residual judge/rubric sensitivity.

| Failure mode | Model-case | Cases | What failed | Typical cases |
|---|---|---|---|---|
| Multi-rule synthesis | 26 | 12 | Retrieves local facts but fails to combine rules, precedence, workflow order, or causal relations. | CPMI stability, CHAPS/CLS fallback, SVF onboarding |
| Scope or exception boundary | 26 | 10 | Misapplies an exclusion, exception, entity role, jurisdictional scope, or transaction-state boundary. | CHATS member exclusion, all-or-none, reporting boundary |
| Over-expanded answer path | 18 | 7 | Adds extra rules, exceptions, or operational actions that change or obscure the requested decision. | Visa OCT, Buna access, PayPal refund |
| Modal or threshold precision | 17 | 10 | Confuses may/must, possible/certain, immediate/conditional, or a numeric/date threshold. | Zelle possible delay, ISO 20022 may need, 21-day conversion |
| Judge/rubric sensitivity | 7 | 5 | The answer is substantially correct, but automatic scoring is sensitive to extra wording or missing exact phrasing. | offshore RMB conversion, same-day bill payment |
| Liability or responsibility boundary | 5 | 3 | Does not cleanly assign responsibility among platforms, banks, providers, or users. | Western Union wallet, BOCHK e-CNY, gateway vs escrow |

Table 16: **Representative diagnostic cases.** Each row connects an observed model–case transition to the concrete payment rule that failed and the resulting triage implication.

| Signal | Model / case | Concrete failure | Triage implication |
|---|---|---|---|
| Context Rescued | Gemini 3.1 Pro / OTC Clear liability | In Closed, the model reverses the source rule by assigning responsibility to clearing members and exempting OTC Clear. With context, it correctly states that OTC Clear is responsible for item accuracy while SI, HKICL, and MA are exempt. | Supply the exact source rule when generic liability priors conflict with scheme-specific allocation. |
| Context-Use Gap | Gemini 3.1 Pro / HKICL all-or-none | The Open answer explains unwind and re-settlement, but incorrectly defines the general-member branch as excluding only HKSCC; the rule also excludes HKCC and SEOCH. | Verify entity scope and exception lists after retrieving the relevant rule. |
| Task-Input Attack | Qwen3 32B / MoneyGram debit timing | The task foregrounds a 4:30 p.m. submission and removes common delay factors. The source says the ACH cut-off does not apply to debit cards and gives no same-day guarantee, yet the model infers same-day debit and gives an internally inconsistent conclusion. | Check rule applicability before applying a familiar threshold or workflow prior. |

## G Human Review Instructions

Human reviewers assessed benchmark validity and judged correctness, not payment, legal, compliance, or customer-facing advice. Review artifacts included the question, supplied source context, reference answer, grading rubric, model output, judge verdict, and source identifiers when available. Reviewers marked whether an item was source-supported, answerable, unambiguous, payment-specific, diagnostically useful, and fairly scorable; Attacked Open review additionally checked that the perturbation preserved the source-supported answer path.

Four practitioners first jointly reviewed 30 calibration cases (10 per subset) to align application of this form. The remaining candidates were assigned in four disjoint subsets for one-practitioner first-pass review, then merged and jointly adjudicated. The form supported ADMIT, HOLD OUT, and REVISE; an item was admitted only when adjudication left no unresolved issue. We define a disputed item operationally as one with an unresolved concern about source support, answer/rubric validity, ambiguity, diagnostic value, duplicate coverage, source stability, or attack validity after discussion. Such items were held out. Cases answered incorrectly by Gemini 3.1 Pro, GPT-5.5, or Claude Opus 4.7 received an additional two-expert validity review, and unresolved item-defect cases were removed.

For the sample-based proxy IAA, 60 candidates (20 per subset) were randomly sampled. Each was re-reviewed by an original payment-domain practitioner who had not conducted its first-pass review and was blind to the original label, using the same criteria. Binary admission agreement was 57/60 ($95.0\%$; Cohen's $\kappa = 0.900$). This is a reliability check of the admission decision on a sample, not full-pool IAA.

Reviewers were informed that all materials were for benchmark construction and research evaluation only, that no private customer records, account data, transaction logs, or personally identifiable information should be used, and that benchmark cases should not be interpreted as operational certification or professional advice.